

LA HW

Included exercises: 2, 4-5, 9, 11, 14, 21-22, 24, 58, 62, 65-67, 71-72, 75, 79, 82

Exercises 1-8

In Exercises 1–8, find a basis for each subspace $\mathcal{S}^\perp$.

$$\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

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Original Exercise 2

$$\mathcal{S} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$$

Original Exercise 4

$$\mathcal{S} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} \right\}$$

Original Exercise 5

$$\mathcal{S} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 3 \\ 2 \end{bmatrix} \right\}$$

Original Exercise 9

Find the distance from $\mathbf{u}$ to $\mathcal{W}$.

$$\mathbf{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \text{and} \quad \mathcal{S} = \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

Original Exercise 11

$$\mathbf{u} = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix} \quad \text{and} \quad \mathcal{S} = \left\{ \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

Original Exercise 14

$$\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ 4 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathcal{S} = \left\{ \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \frac{1}{2} \begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Original Exercise 21

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 6 \end{bmatrix} \text{ and } W = \text{Col} \begin{bmatrix} 1 & 1 & 5 \\ -1 & 2 & 1 \\ -1 & 1 & -1 \\ 2 & -1 & 4 \end{bmatrix}$$

Original Exercise 22

$$\mathbf{u} = \begin{bmatrix} -3 \\ 7 \\ -1 \\ 5 \end{bmatrix} \text{ and } W \text{ is the solution set of}$$

$$x_1 - x_2 + 2x_3 + x_4 = 0.$$

Original Exercise 24

$$\mathbf{u} = \begin{bmatrix} 7 \\ 4 \\ 1 \\ 2 \end{bmatrix} \text{ and } W = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 2 \\ 2 \end{bmatrix} \right\}$$

Exercise 58

60. Prove that for any subspace W of $\mathcal{R}^n$, $(W^\perp)^\perp = W$.

Original Exercise 58

Let W be a subspace of $\mathcal{R}^n$, and let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases for W and $W^\perp$, respectively. Apply the orthogonal decomposition theorem to prove that

(a) $\mathcal{B}_1 \cup \mathcal{B}_2$ is a basis for $\mathcal{R}^n$.

Original Exercise 62

$$(b) (\text{Col } A)^\perp = \text{Null } A^T$$

Prove that if $\mathcal{S}_1$ and $\mathcal{S}_2$ are subsets of $\mathcal{R}^n$ such that $\mathcal{S}_1$ is

Original Exercise 65

rows constitute a basis for W .)

Let A be an $n \times n$ matrix. Prove that if $\mathbf{v}$ is a vector in

Original Exercise 66

Let V and W be subspaces of $\mathcal{R}^n$ such that every vector in V is orthogonal to every vector in W . Prove that

Original Exercise 67

$$\dim V + \dim W \leq n.$$

Let W be a subspace of $\mathcal{R}^n$.

$$(a) \text{ Prove that } (P_W)^2 = P_W.$$

Original Exercise 71

Let W be a subspace of $\mathcal{R}^n$. Prove that $(P_W \mathbf{u}) \cdot \mathbf{v} =$

Original Exercise 72

Let W be a subspace of $\mathcal{R}^n$. Prove that $P_W P_{W^\perp} =$

Original Exercise 75

Describe the subspace Z of $\mathcal{R}^n$ such that $P_Z = P_V + P_W$.

Suppose that $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is an orthonormal basis for a subspace W of $\mathcal{R}^n$. Let C be the $n \times k$ matrix whose columns are the vectors in $\mathcal{B}$. Prove the following statements:

(a) $C^T C = I_k$.

Original Exercise 79

Obtain a basis for $V^\perp$ as in Example 2 of Section 6.2.

(a) Let $W = \text{Null } C$, where C is an $m \times n$ matrix of rank m . Prove that

$$P_{W^\perp} = C^T (CC^T)^{-1} C$$

and

$$P_W = I_n - C^T (CC^T)^{-1} C.$$

Original Exercise 82

Use the method described in Exercise 77 to find P_W .

Suppose that P is an $n \times n$ matrix such that $P^2 = P^T = P$. Prove that P is the orthogonal projection matrix P_W for some subspace W of $\mathcal{R}^n$.